Quantum-Enhanced Tacit Coordination for Latency-Constrained Trading
A Game-Theoretic, Bayesian, and Microstructure-Based Case for Profitability, Implementability, and Strategic Benefit

Clete Roberts

March 2026

Contents

1 Executive Summary **7**

2 Introduction and Strategic Motivation **8**

3 Literature Context **8**

4 A Better Trading Problem **9**

4.1 Why the Toy Example Should Be Replaced 9

4.2 The Cross-Venue Toxicity-Confirmation Game 9

4.3 A More Realistic Objective 10

5 Formal Team-Game Model **11**

5.1 The Bayesian Team View 11

5.2 Posterior-Threshold Structure 11

6 XOR Reduction and the CHSH Structure **12**

7 From Coordination Accuracy to Profit **13**

7.1 Interpretation 13

8 A Microstructure Model for $\bar{\Delta}$ **14**

8.1 Adverse Selection 14

8.2 Inventory Dislocation 14

8.3 Missed Opportunity Cost . . . 14
8.4 Putting the Pieces Together . . . 15

9 General Bayesian Signal Model **15**
9.1 Latent-State Model . . . 15
9.2 Conditional Probabilities . . . 15
9.3 Threshold Choice . . . 16

10 Bayesian Game-Theoretic Extensions **16**
10.1 Heterogeneous Costs Across States . . . 16
10.2 Correlated Private Signals . . . 16
10.3 Unequal Venue Importance . . . 16
10.4 Dynamic Bayesian Teams . . . 17

11 Robustness to Modeling Error **17**

12 Implementation Architecture **18**
12.1 The Right Way to Frame Implementation . . . 18
12.2 Two Physical Architectures . . . 18
12.3 Hybrid Quantum-Classical Control . . . 18
12.4 Integration into Existing Systems . . . 19

13 Classical Baselines **19**
13.1 Deterministic Fallback Policy . . . 19
13.2 Shared-Randomness Policy . . . 19
13.3 Cheap-Talk or Delayed Communication Benchmark . . . 19
13.4 Predictive Local Approximation . . . 19

14 Multi-Venue Generalization **20**
14.1 Problem Statement . . . 20
14.2 Threshold Game . . . 20
14.3 Parity Encoding . . . 20
14.4 Economic Interpretation . . . 20

15 A Weighted Multi-Venue Profit Formula **21**

16 Factor-Model Calibration **21**

16.1 Why Factor Calibration Matters . . . 21
16.2 Latent Factor Specification . . . 21
16.3 Binary Reduction . . . 22
16.4 Calibrating $\bar{\Delta}$ . . . 22

17 Simulation Framework **22**
17.1 Event-Driven Monte Carlo . . . 22
17.2 Suggested Parametric Baseline . . . 23
17.3 Performance Metrics . . . 23

18 Risk-Adjusted Value and Governance **23**
18.1 Risk-Adjusted Objective . . . 23
18.2 Operational Controls . . . 24

19 Why the Company Could Rationally Care **24**

20 Numerical Illustration **24**

21 Empirical Definition of Coordination Events **25**
21.1 Local Signal Structure . . . 25
21.2 Coordination Requirement . . . 26
21.3 Formal Event Definition . . . 26

22 Economic Cost of Miscoordination **26**
22.1 Execution Loss Model . . . 27
22.2 Expected Loss . . . 27

23 Classical Baseline Coordination Performance **27**
23.1 Shared Randomness Strategy . . . 27
23.2 Observed Desk Performance . . . 28

24 Profitability Model **28**
24.1 Break-Even Condition . . . 28

25 Operational Feasibility at Production Latency **29**
25.1 Latency Decomposition . . . 29

26 Mapping XOR Coordination Games to Trading Decisions **29**

26.1 Game Variables . . . 29
26.2 Trading Interpretation . . . 30

27 Empirical Backtesting Framework **30**
27.1 Backtest Steps . . . 30

28 Empirical Calibration Using Historical Market Data **30**
28.1 Data Sources . . . 31
28.2 Signal Estimation . . . 31
28.3 Event Frequency Estimation . . . 31

29 Monte Carlo Microstructure Simulation **32**
29.1 Price Dynamics . . . 32
29.2 Order Flow Model . . . 32
29.3 Coordination Policies . . . 32
29.4 Simulation Output . . . 32

30 Multi-Venue Coordination Extension **33**
30.1 Generalized Coordination Probability . . . 33
30.2 Economic Scaling . . . 33

31 What a Bulletproof Version Can and Cannot Mean **34**

32 Expanded Theoretical Results **34**
32.1 Nonuniform Input Distribution . . . 34
32.2 State-Weighted Optimization . . . 35
32.3 Approximate Reduction Error . . . 35

33 A Dynamic Inventory-Control Extension **35**

34 A Three-Posture Generalization **36**

35 Sensitivity Tables to Include in a Final Memo **36**

36 Model-Risk Appendix Summary **37**

37 Conclusion **37**

38 Semidefinite Optimization View of the Quantum Problem **37**
38.1 Weighted XOR Objective . . . 38
38.2 Why This Helps Internally . . . 38

39 Signal Engineering for a Trading Desk **38**
39.1 Queue-Pressure Signals . . . 38
39.2 Trade-Flow Signals . . . 39
39.3 Crossed-Book and Fragmentation Signals . . . 39
39.4 Event Compression into a Binary Signal . . . 39

40 Market-Impact and Execution-Cost Extension **39**
40.1 Temporary Impact . . . 39
40.2 Permanent Impact and Strategic Selectivity . . . 40

41 Portfolio-Level Extension **40**
41.1 Vector Inventory State . . . 40
41.2 Basket Event Example . . . 40

42 Backtest Design for a Production Research Pipeline **41**
42.1 Event Definition . . . 41
42.2 Counterfactual Labeling . . . 41
42.3 Avoiding Look-Ahead Bias . . . 41
42.4 Cross-Validation by Regime . . . 41

43 Scenario Analysis and Stress Tests **42**
43.1 Lower Entanglement Availability . . . 42
43.2 Signal Deterioration . . . 42
43.3 Higher Cost of Switching . . . 42
43.4 Latency Compression by Classical Infrastructure . . . 42

44 A Research Roadmap for the Company **43**
44.1 Stage 1: Offline Quant Research . . . 43
44.2 Stage 2: Historical Replay with Simulated Coordination Oracle . . . 43
44.3 Stage 3: Shadow Mode . . . 43
44.4 Stage 4: Bounded-Risk Pilot . . . 43

45 Communication Strategy for Internal Stakeholders **43**
45.1 For Traders . . . 43
45.2 For Quant Researchers . . . 44
45.3 For Technology Leads . . . 44
45.4 For Risk and Compliance . . . 44

46 Further Mathematical Appendices **44**
46.1 A Concentration Bound for Estimated Value . . . 44
46.2 A Conservative Lower Bound on Net Value . . . 44
46.3 Model Averaging Across Regimes . . . 45

47 Extended Conclusion **45**

48 Additional Calibration and Validation Considerations **45**
48.1 Estimating the Relevant Event Rate . . . 45
48.2 Out-of-Sample Penalty Stability . . . 46
48.3 Conservative Event Filtering . . . 46

49 Detailed Dynamic-Programming Interpretation **46**
49.1 Approximation Gap . . . 47

50 Suggested Tables for the Final Submission **47**
50.1 Table 1: Event Statistics . . . 47
50.2 Table 2: Policy Comparison . . . 47
50.3 Table 3: Break-Even Cost Frontier . . . 47

51 Final Managerial Framing **47**

A Notation Summary **48**

B Exact Enumeration of Deterministic Binary Local Policies **48**

C Proof of the Break-Even Condition **48**

D Pseudo-Code for Historical Backtest Calibration **49**

E Suggested Overleaf Compilation Notes **49**

Abstract

This paper develops a mathematically rigorous and practically oriented replacement for the high-frequency trading example proposed in *Coordinating Decisions via Quantum Telepathy*. The central idea is to recast a latency-constrained multi-venue trading decision as a cooperative Bayesian game with local observations, no real-time communication, and a joint profit function. The resulting optimization problem is a tacit coordination problem in the sense of Ding and Jiang. We show how a realistic cross-venue toxicity-confirmation problem naturally reduces to an XOR-type coordination game, derive the associated classical and quantum values, and map the coordination gap into dollars using adverse-selection, inventory, and opportunity-cost primitives. We then extend the model in five directions that are important for a real trading firm: heterogeneous venue quality, imperfect and correlated private signals, multi-venue coordination, factor-based calibration, and Monte Carlo simulation. Finally, we derive explicit implementation and break-even conditions that connect trading frequency, average coordination-error loss, and deployment cost. The resulting document is written as a persuasive internal research memorandum: its purpose is not to claim that every firm will necessarily profit, but rather to identify a sharp set of sufficient conditions under which the proposed coordination layer is value-accretive and operationally defensible.

1 Executive Summary

The motivating paper of Ding and Jiang introduces a broad framework for *tacit coordination problems*: situations in which multiple agents must coordinate actions on the basis of local observations while lacking time to communicate before acting. Their Appendix A formalizes the problem abstractly, and Section 3 applies it to a stylized HFT setting. That starting point is valuable, but the particular HFT example is too toy-like to persuade most market practitioners. The natural question is therefore not whether the framework is interesting; it is whether one can build a better trading example that is realistic, mathematically clean, and economically compelling.

This document answers that question in the affirmative. The core contribution is a new market-microstructure game:

> Two or more trading engines, colocated at different venues, each observe a binary local signal about sweep risk, informed-flow toxicity, or short-horizon directional pressure. The firm must determine a joint posture—protective, passive, aggressive, or directional—before communication is possible. The wrong posture creates measurable losses through adverse selection, inventory dislocation, and missed opportunities. The right posture depends on the *joint* signal, not on any one local signal in isolation.

In this setting, classical no-communication policies are constrained by Bell-type bounds, while entanglement-assisted policies can achieve a strictly larger feasible set of correlations. We convert that abstract advantage into the language a company cares about:

1. a precise team-game objective,
2. a tractable implementation architecture,
3. a calibration methodology on historical data,

4. a simulation methodology, and
5. a break-even inequality linking the coordination advantage to profit.

The final recommendation is not "3replace trading logic with quantum hardware."4 It is to treat entanglement-assisted coordination as a *narrow decision layer* that can be inserted into an otherwise classical low-latency stack. The only events that call the coordination primitive are those where the firm already faces costly no-communication miscoordination. This narrower framing is both more realistic and more defensible.

2 Introduction and Strategic Motivation

Trading firms spend heavily to reduce latency because speed creates edge. Yet speed also produces a paradox: when decision windows become shorter than communication delays, the firm loses the ability to coordinate distributed actions in real time. At that point, optimization is no longer primarily a matter of signal quality or execution speed; it becomes a matter of distributed decision theory.

The economic relevance of this phenomenon is easy to see. Suppose a market-making firm quotes the same name on multiple venues. A local engine on Venue 1 sees queue depletion, hidden-liquidity signatures, and aggressive marketable order flow that suggest a toxic sweep. A local engine on Venue 2 sees a weaker but related warning. If both venues are jointly signaling the same latent event, the firm should immediately widen, cancel, skew, or switch posture. If not, it should remain passive and continue collecting spread. Because the decision must occur in microseconds, the engines may not have time to coordinate through ordinary communication. The cost of getting the joint action wrong is not theoretical: it shows up in markouts, fill quality, inventory imbalance, and missed spread capture.

This paper therefore frames the company problem in three layers.

1. **Decision-theoretic layer.** The firm faces a cooperative Bayesian game with local private information and no communication.
2. **Microstructure layer.** The joint action controls adverse selection, inventory, and opportunity capture.
3. **Implementation layer.** The coordination primitive must fit inside a low-latency architecture with bounded cost and bounded operational risk.

A proposal will only persuade the company if it succeeds on all three layers. A mathematically elegant result without an implementation path is not enough. Likewise, a hardware story without an economic theorem is not enough. The purpose of this document is to bridge those gaps.

3 Literature Context

Bell's 1964 theorem established that no local hidden-variable theory can reproduce all quantum correlations. Clauser, Horne, Shimony, and Holt later gave the CHSH formulation that remains the

canonical operational inequality for two-party nonlocal games. Tsirelson derived the sharp quantum bound for CHSH, now called Tsirelson's bound. In the language of computer science and game theory, this advantage appears as *pseudo-telepathy*: players sharing entanglement can coordinate outputs in ways impossible with purely classical shared randomness.

Ding and Jiang take this body of theory and ask a practical question: when can such quantum correlations help with real-world coordination tasks? Their answer is to define tacit coordination problems and show that Bell-type quantum advantages can arise whenever multiple agents make local observations and cannot communicate before choosing actions. Their HFT section is important because it grounds the framework in latency-sensitive markets, where propagation delays are meaningful at decision timescales. However, their stylized trading example is only a first pass. This paper keeps their framework but changes the economics.

There are additional literatures relevant to the proposal. The first is Bayesian team theory, in which multiple decision makers share a common objective but act on private information. The second is market microstructure, especially adverse selection in market making and the economics of short-horizon toxicity detection. The third is decentralized stochastic control, where local controllers must coordinate under information constraints. A persuasive company memo should make clear that the proposed strategy is not merely "3quantum for quantum's sake."4 It is a new solution concept for an old decentralized-control problem.

4 A Better Trading Problem

4.1 Why the Toy Example Should Be Replaced

A trader is unlikely to get excited by a correlation-flip story unless the path from correlation changes to dollars is immediate and measurable. What traders do care about are phenomena with direct P&L consequences:

- toxic sweeps,
- informed-flow detection,
- queue-jump risk,
- inventory dislocation,
- stale quoting under cross-venue pressure,
- and the speed tradeoff between staying passive and turning protective.

These are the places where coordination errors already cost money. Any new strategy should therefore target them directly.

4.2 The Cross-Venue Toxicity-Confirmation Game

Consider two venues, indexed by $j \in \{1, 2\}$. At each decision epoch t, venue j produces a local binary signal

$$I_{j,t} \in \{0, 1\},$$

where $I_{j,t} = 1$ means “3my local order book and recent trade flow indicate elevated toxicity or directional pressure,“4 and $I_{j,t} = 0$ means “3no such warning.“4

Each venue engine chooses a local binary action

$$A_{j,t} \in \{0, 1\}.$$

The firm-level posture is determined by the parity or, more generally, by a low-dimensional map from $(A_{1,t}, A_{2,t})$ into a set of postures such as passive, protective, or directional. The minimal case is an XOR posture rule:

$$P_t = A_{1,t} \oplus A_{2,t},$$

with one parity corresponding to a hedged/passive stance and the other parity corresponding to a protective/directional stance.

The economic intuition is simple: one local warning is often too noisy to justify a full posture change, but double confirmation across venues is materially stronger evidence of a latent short-horizon event. If communication were free, the firm would choose the globally optimal posture as a deterministic function of $(I_{1,t}, I_{2,t})$. Under latency constraints, the engines must act using only local observations and any pre-shared resource.

4.3 A More Realistic Objective

Let $E_t \in \{0, 1\}$ be the latent event that there truly exists a short-horizon directional or toxicity shock. Let $P_t \in \{H, D\}$ denote the firm posture: H for hedged/passive and D for defensive or directional. The one-shot payoff is

$$\Pi_t(P_t, E_t) = \begin{cases} \pi_D, & P_t = D,\ E_t = 1, \\ -\ell_D, & P_t = D,\ E_t = 0, \\ \pi_H, & P_t = H,\ E_t = 0, \\ -\ell_H, & P_t = H,\ E_t = 1. \end{cases}$$

Here:

- $\pi_D > 0$ is the gain from switching posture when the latent event is real,
- $\ell_D > 0$ is the false-positive cost of switching unnecessarily,
- $\pi_H > 0$ is the gain from remaining passive when conditions are normal,
- $\ell_H > 0$ is the loss from failing to switch when the event is real.

This payoff structure converts the problem into the language of expected P&L. A company can debate the correct calibration of the four parameters, but the structure itself is standard and economically interpretable.

5 Formal Team-Game Model

Definition 5.1 (Two-player tacit coordination game)**.** A two-player tacit coordination game is a tuple

$$\mathcal{G} = (\mathcal{I}_1, \mathcal{I}_2, \mathcal{A}_1, \mathcal{A}_2, \pi, U),$$

where $\mathcal{I}_j$ are local observation spaces, $\mathcal{A}_j$ are local action spaces, π is a joint distribution on $\mathcal{I}_1 \times \mathcal{I}_2$, and U is a common utility function on $\mathcal{I}_1 \times \mathcal{I}_2 \times \mathcal{A}_1 \times \mathcal{A}_2$.

We adopt the two-player case for concreteness, then extend to N venues in Section 14. A classical strategy is a pair of local decision rules

$$f_j : \mathcal{I}_j \to \mathcal{A}_j,$$

possibly randomized via shared randomness. A quantum strategy is a shared entangled state together with local measurements chosen as functions of the local observations.

5.1 The Bayesian Team View

Let the private signals I_1, I_2 be generated conditionally on a latent state Θ taking values in a finite or continuous space. The common prior is μ and the signal channel is $p(i_1, i_2 \mid \theta)$. The firm's expected value under local policies σ_1, σ_2 is

$$V(\sigma_1, \sigma_2) = \mathbb{E}\big[U(\Theta, \sigma_1(I_1), \sigma_2(I_2))\big].$$

This is a cooperative Bayesian game: the agents are not strategic adversaries; they are decentralized controllers with a shared objective. A practical benefit of this formulation is that it allows direct comparison with the literature on team decision theory and decentralized control.

5.2 Posterior-Threshold Structure

Let

$$q(i_1, i_2) = \mathbb{P}(E = 1 \mid I_1 = i_1, I_2 = i_2).$$

The expected payoff of choosing posture D is

$$V_D(i_1, i_2) = q(i_1, i_2)\pi_D - (1 - q(i_1, i_2))\ell_D,$$

while the expected payoff of choosing posture H is

$$V_H(i_1, i_2) = (1 - q(i_1, i_2))\pi_H - q(i_1, i_2)\ell_H.$$

The firm prefers D iff $V_D(i_1, i_2) > V_H(i_1, i_2)$, i.e.

$$q(i_1, i_2) > \tau := \frac{\pi_H + \ell_D}{\pi_D + \ell_H + \pi_H + \ell_D}.$$

Thus the globally optimal rule is a posterior threshold rule. This is important for persuasion because it grounds the proposed posture logic in ordinary Bayesian economics rather than in quantum rhetoric.

6 XOR Reduction and the CHSH Structure

The simplest and cleanest reduced-form version is obtained when the threshold rule is crossed only in the double-confirmation state $(1, 1)$. Then the globally optimal posture is

$$P^*(i_1, i_2) = \begin{cases} D, & (i_1, i_2) = (1, 1), \\ H, & \text{otherwise.} \end{cases}$$

Coding postures through parity gives

$$A_1 \oplus A_2 = 1 - I_1 I_2.$$

This is exactly a CHSH-type winning condition up to relabeling. The reduction matters for two reasons.

1. It makes the classical and quantum values analytically transparent.
2. It shows that the nonlocal-game advantage is not bolted on artificially; it emerges from a natural posterior-threshold trading rule.

Proposition 6.1 (Classical value for the symmetric XOR reduction)**.** *Assume (I_1, I_2) are uniformly distributed on $\{0, 1\}^2$ and the payoff is one for satisfying the parity condition $A_1 \oplus A_2 = 1 - I_1 I_2$ and zero otherwise. Then the optimal classical value is*

$$\omega_C = \frac{3}{4}.$$

Proof. For deterministic local rules, the four parity constraints cannot all be satisfied simultaneously because their product is inconsistent. Hence at most three of the four input states can be won. Shared randomness can only convexify deterministic strategies and therefore cannot exceed $3/4$. □

Proposition 6.2 (Quantum value for the symmetric XOR reduction)**.** *For the same game, there exists an entanglement-assisted strategy achieving*

$$\omega_Q = \cos^2(\pi/8) \approx 0.8536.$$

Proof. Use a maximally entangled Bell state and choose local projective measurements with relative angles corresponding to the CHSH optimum. The induced correlation achieves Tsirelson's bound $2\sqrt{2}$ on the CHSH expression, which translates into success probability $\cos^2(\pi/8)$. □

Remark 6.3. The value gap

$$\Delta\omega = \omega_Q - \omega_C \approx 0.1036$$

is the fundamental coordination gain available in the reduced symmetric game. The business case is to multiply this gap by the dollars lost when the firm coordinates incorrectly.

7 From Coordination Accuracy to Profit

The strongest mathematical selling point for the company is not merely that $\omega_Q > \omega_C$. It is that expected profit is an affine function of the correctness probability whenever the underlying business problem is a binary posture problem with a wrong-action penalty.

Define the state-dependent dollar loss from incorrect posture by

$$\Delta(i_1, i_2) := \mathbb{E}[\Pi(P^*(i_1, i_2), E) \mid i_1, i_2] - \mathbb{E}[\Pi(P^{\text{wrong}}(i_1, i_2), E) \mid i_1, i_2].$$

If a strategy class X achieves state-dependent correctness probabilities $s_X(i_1, i_2)$, then

$$\mathbb{E}[\Pi_X] = \Pi_{\text{base}} + \sum_{i_1, i_2} \pi(i_1, i_2) s_X(i_1, i_2) \Delta(i_1, i_2).$$

Consequently,

$$\mathbb{E}[\Pi_Q] - \mathbb{E}[\Pi_C] = \sum_{i_1, i_2} \pi(i_1, i_2)\big(s_Q(i_1, i_2) - s_C(i_1, i_2)\big)\Delta(i_1, i_2).$$

In the symmetric-loss reduced game where $\Delta(i_1, i_2) = \bar{\Delta}$ for all states,

$$\mathbb{E}[\Pi_Q] - \mathbb{E}[\Pi_C] = (\omega_Q - \omega_C)\bar{\Delta}.$$

This yields an immediate annualized formula.

Theorem 7.1 (Annual incremental gross profit)**.** *Suppose the relevant decision occurs N times per year and the average dollar penalty of a coordination error is $\bar{\Delta}$. In the symmetric reduced game,*

$$\Delta\Pi_{\text{gross}} = N(\omega_Q - \omega_C)\bar{\Delta}.$$

Proof. Apply linearity of expectation across independent or merely stationary decision epochs and use the previous identity. □

7.1 Interpretation

The theorem isolates the three things the company needs to assess:

1. **Frequency** N: how often does the costly no-communication decision occur?
2. **Gap** $\omega_Q - \omega_C$: what improvement is available in the appropriate reduced game?
3. **Penalty** $\bar{\Delta}$: how much money is lost when the firm coordinates incorrectly?

Each quantity is measurable or at least estimable from historical market data, risk records, and implementation design. That is why the formula is persuasive.

8 A Microstructure Model for $\bar{\Delta}$

To move from qualitative claims to calibration, we need an explicit decomposition of the miscoordination penalty. Let a wrong posture cost the firm through three channels:

$$\bar{\Delta} = \bar{\Delta}_{\text{AS}} + \bar{\Delta}_{\text{INV}} + \bar{\Delta}_{\text{MISS}},$$

where:

- $\bar{\Delta}_{\text{AS}}$ is adverse-selection loss,
- $\bar{\Delta}_{\text{INV}}$ is inventory dislocation cost,
- $\bar{\Delta}_{\text{MISS}}$ is opportunity cost from failing to switch when the latent event is present.

8.1 Adverse Selection

If the firm remains passive when informed flow is present, it may be picked off. Denote by Q the executed size, M the post-fill markout over a short horizon, and c any spread capture retained after the event. Then a simple representation is

$$\Delta_{\text{AS}} = Q(M - c)_+,$$

where $(x)_+ = \max\{x, 0\}$. Taking expectations over the subset of events where the wrong passive posture is chosen gives

$$\bar{\Delta}_{\text{AS}} = \mathbb{E}\big[Q(M - c)_+ \mid \text{wrong passive posture}\big].$$

8.2 Inventory Dislocation

Suppose a wrong joint action creates inventory vector $X \in \mathbb{R}^d$ over venues or instruments. A common reduced-form penalty is quadratic:

$$\Delta_{\text{INV}} = X^{\mathsf{T}} \Lambda X,$$

where Λ is a positive semidefinite risk-cost matrix encoding temporary impact, inventory aversion, and expected unwinding cost. The average inventory penalty under miscoordination is then

$$\bar{\Delta}_{\text{INV}} = \mathbb{E}[X^{\mathsf{T}} \Lambda X \mid \text{miscoordination}].$$

8.3 Missed Opportunity Cost

If the firm remains hedged when it should have switched posture, it forgoes expected alpha or avoids a protective action too slowly. Let α be the expected per-share edge conditional on the latent event and Q^* the optimal response size. Then

$$\Delta_{\text{MISS}} = Q^* \alpha,$$

so

$$\bar{\Delta}_{\text{MISS}} = \mathbb{E}[Q^* \alpha \mid \text{wrong hedged posture}].$$

8.4 Putting the Pieces Together

A company memo becomes materially stronger when it says not only "3coordination errors cost money,"4 but exactly *how*. The decomposition above allows calibration from standard HFT analytics:

- markout studies for adverse selection,
- inventory slippage and impact models for risk cost,
- event-conditioned alpha capture for missed opportunities.

9 General Bayesian Signal Model

The binary reduced form is analytically convenient, but a production-caliber argument should show how it emerges from richer signals.

9.1 Latent-State Model

Let the latent state satisfy

$$\Theta_t = \beta^\mathsf{T} F_t + \eta_t,$$

where $F_t \in \mathbb{R}^k$ is a vector of common factors, $\beta \in \mathbb{R}^k$ is a loading vector, and η_t is idiosyncratic noise. The factors may include broad-market stress, ETF flow pressure, volatility shocks, macro-event dummies, or sector-specific flow indicators.

At venue j, define a continuous microstructure score

$$S_{j,t} = \gamma_j \Theta_t + \varepsilon_{j,t},$$

where γ_j is venue-specific sensitivity and $\varepsilon_{j,t}$ is local noise. The binary signal is thresholded:

$$I_{j,t} = \mathbf{1}\{S_{j,t} > \tau_j\}.$$

This setup yields a full probabilistic model for $\pi(i_1, i_2)$ and $q(i_1, i_2)$.

9.2 Conditional Probabilities

When $(\eta_t, \varepsilon_{1,t}, \varepsilon_{2,t})$ is jointly Gaussian, the probabilities of each binary signal pattern can be written in terms of multivariate normal orthant probabilities. In practical calibration, these can be estimated either parametrically or by empirical binning on historical data.

9.3 Threshold Choice

The thresholds τ_j should not be chosen arbitrarily. A useful calibration criterion is to choose them so that each local signal has a target false-positive rate under benign conditions while preserving enough sensitivity under the latent event. A more sophisticated criterion is to optimize thresholds jointly with posture cost parameters to maximize out-of-sample expected profit.

10 Bayesian Game-Theoretic Extensions

10.1 Heterogeneous Costs Across States

The reduced symmetric game assumes a common wrong-action cost $\bar{\Delta}$. In practice, the penalty depends on the state. Define

$$\Delta_{00},\ \Delta_{01},\ \Delta_{10},\ \Delta_{11}$$

for the four signal states. Then

$$\mathbb{E}[\Pi_Q] - \mathbb{E}[\Pi_C] = \sum_{i_1,i_2\in\{0,1\}} \pi_{i_1 i_2}(s^Q_{i_1 i_2} - s^C_{i_1 i_2})\Delta_{i_1 i_2}.$$

A particularly favorable environment for the company is one in which Δ_{11} is much larger than the other penalties, because the entire economic point of the strategy is to resolve the globally informative double-confirmation state more accurately.

10.2 Correlated Private Signals

Signals need not be conditionally independent. Suppose

$$\mathrm{Corr}(\varepsilon_{1,t}, \varepsilon_{2,t}) = \rho_\varepsilon.$$

High positive local-noise correlation can either help or hurt the business case. It can help because it increases the incidence of useful joint confirmations under a real common shock. It can hurt because it also increases the incidence of false joint warnings if local noise co-moves for non-informational reasons. This is precisely why calibration is important: the economic value depends on the ratio of true and false joint confirmations, not just on raw correlation.

10.3 Unequal Venue Importance

Some venues matter more than others. Let venue-specific weights w_j encode contribution to expected fill, inventory exposure, or information quality. A weighted posture objective may be written as

$$U = \sum_{j=1}^{2} w_j u_j(I_1, I_2, A_1, A_2),$$

with $w_1 + w_2 = 1$. The reduction to an XOR game still applies whenever the optimal action depends only on a parity relation after relabeling, but the penalty term becomes weighted.

10.4 Dynamic Bayesian Teams

A realistic firm does not solve a one-shot problem; it solves a sequence of such problems. Let inventory state X_t evolve as

$$X_{t+1} = X_t + B(A_{1,t}, A_{2,t}, \xi_t),$$

where ξ_t summarizes fills and exogenous flow. The objective is to maximize

$$\mathbb{E}\left[\sum_{t=0}^{T-1} \delta^t (\Pi_t - X_t^\top \Lambda X_t)\right],$$

for discount factor $\delta \in (0, 1]$. In this dynamic setting, the one-shot coordination layer remains useful provided it improves the expected stage payoff net of the marginal effect on future inventory risk.

11 Robustness to Modeling Error

The company will ask not only whether the idea works under one exact specification, but whether it is fragile. The right response is a collection of robustness propositions.

Proposition 11.1 (Monotonicity in coordination penalty)**.** *For fixed N, implementation cost, and classical/quantum values, the net value of adoption is increasing in the average coordination-error penalty $\bar{\Delta}$.*

Proof. Immediate from linearity of the annual gross profit formula. □

Proposition 11.2 (Monotonicity in event frequency)**.** *For fixed $\bar{\Delta}$ and costs, net value is increasing in the annual frequency N of relevant decision epochs.*

Proof. Immediate from the same formula. □

Proposition 11.3 (Break-even threshold)**.** *Let K_{fixed} be annualized fixed deployment cost and k_{var} variable cost per event. The strategy is profitable if and only if*

$$\bar{\Delta} > \frac{K_{\text{fixed}}/N + k_{\text{var}}}{\omega_Q - \omega_C}.$$

Proof. Net annual value is

$$\Delta\Pi_{\text{net}} = N(\omega_Q - \omega_C)\bar{\Delta} - K_{\text{fixed}} - N k_{\text{var}}.$$

Rearrange the positivity condition. □

Remark 11.4. This is the central capital-allocation inequality. It converts the entire proposal into a measurable threshold test.

12 Implementation Architecture

12.1 The Right Way to Frame Implementation

The implementation question should not be framed as "3replace the trading system with quantum hardware."4 That framing is guaranteed to fail. The right framing is:

> Can the firm insert a narrow entanglement-assisted correlation layer into a classical low-latency stack, such that the layer is invoked only on decision epochs where no-communication miscoordination is already costly?

This narrower framing has three advantages. First, it reduces the required event rate. Second, it constrains operational scope. Third, it makes fallback-to-classical logic straightforward.

12.2 Two Physical Architectures

There are two conceptually clean architectures.

1. **Direct photonic link.** An entangled-photon source distributes one half of each pair to each site. The local signal selects the measurement basis, the detector outcome is mapped to a local action bit, and the posture follows from the joint parity relation.
2. **Heralded-memory architecture.** Entanglement is generated and stored in quantum memories, with a heralding signal indicating availability. If entanglement is unavailable for a given epoch, the sites revert to the optimal classical fallback strategy.

For a company audience, the second architecture is usually more persuasive because it makes fault handling explicit.

12.3 Hybrid Quantum-Classical Control

Let $Q_t \in \{0, 1\}$ indicate whether entanglement-assisted coordination is available at time t. Then the effective strategy is

$$\sigma_t = Q_t \sigma_t^{(Q)} + (1 - Q_t)\sigma_t^{(C)}.$$

Expected coordination value is therefore

$$V_{\text{hyb}} = p_Q V_Q + (1 - p_Q)V_C,$$

where $p_Q = \mathbb{P}(Q_t = 1)$. The expected annual gross profit of the hybrid layer becomes

$$\Delta\Pi_{\text{hyb,gross}} = N p_Q(\omega_Q - \omega_C)\bar{\Delta}.$$

This formula is extremely useful: it tells the firm that even partial availability can create value, as long as the economics are favorable.

12.4 Integration into Existing Systems

The rest of the trading stack remains classical:

- market data normalization,
- local feature extraction,
- risk checks,
- order management and throttles,
- kill switches and inventory controls.

The coordination layer only determines a compact posture variable such as “3stay passive,“4 “3turn protective,“4 or “3switch directional.“4 This helps with governance because risk officers can understand exactly what the layer controls.

13 Classical Baselines

A proposal is only persuasive if it is benchmarked against strong classical alternatives. The relevant baselines are not weak heuristics; they are the best things a sophisticated firm would actually do without entanglement.

13.1 Deterministic Fallback Policy

In the reduced CHSH game, one optimal deterministic policy is to always choose the same parity, which wins three of four states. In microstructure terms, this is equivalent to a conservative default posture.

13.2 Shared-Randomness Policy

The firm can also pre-share randomness and randomize across deterministic rules. In the CHSH reduction, this convexifies the deterministic region but cannot exceed the classical value.

13.3 Cheap-Talk or Delayed Communication Benchmark

When the firm can communicate with a slight delay, it may use a lagged correction. This can improve downstream inventory management but does not solve the instantaneous coordination problem. That is exactly why the no-communication decision layer remains relevant.

13.4 Predictive Local Approximation

A strong classical baseline is to have each local engine use a predictive model for the other venue’s likely state. This may work well when the venues are tightly linked, but it is still a local predictor

rather than a true no-communication coordination resource. It should be included in simulation as a baseline, but it does not invalidate the theoretical nonlocal advantage.

14 Multi-Venue Generalization

Most firms operate across more than two venues. We now extend the setup to $m \geq 2$ venues.

14.1 Problem Statement

Each venue $j \in \{1, \ldots, m\}$ observes a binary local signal $I_j \in \{0, 1\}$ and chooses a local binary action $A_j \in \{0, 1\}$. A firm-level posture is generated by a function of the local actions. The natural generalization of the double-confirmation rule is to let the risky posture be triggered only under sufficiently strong multi-venue confirmation, for example when

$$\sum_{j=1}^{m} I_j \geq r$$

for some threshold r.

14.2 Threshold Game

A simple and economically meaningful threshold game is:

$$P^*(I_1, \ldots, I_m) = \begin{cases} D, & \sum_{j=1}^{m} I_j \geq r, \\ H, & \text{otherwise.} \end{cases}$$

For $r = m$, the posture changes only under unanimous confirmation. For lower r, the rule is more aggressive and more frequent.

14.3 Parity Encoding

One can encode the posture via the parity of local actions,

$$A_1 \oplus \cdots \oplus A_m = g(I_1, \ldots, I_m),$$

for a suitable Boolean function g. The resulting game is a generalized XOR game. The exact classical and quantum values depend on the chosen Boolean structure and on the distribution of local inputs.

14.4 Economic Interpretation

The threshold r is not only a mathematical parameter; it is a risk appetite control. A higher r makes the posture switch rarer but usually more accurate. A lower r increases sensitivity but may

generate more false positives. In an internal memo, this creates a natural lever: the company can tune aggressiveness while preserving the same implementation architecture.

15 A Weighted Multi-Venue Profit Formula

Suppose miscoordination at venue j creates expected penalty Δ_j and the probability that venue j is relevant at a given epoch is ν_j. Then a weighted approximation to the annual gross gain is

$$\Delta\Pi^{(m)}_{\text{gross}} = N \sum_{j=1}^{m} \nu_j \kappa_j \Delta_j,$$

where κ_j is the incremental correctness gain attributable to improved coordination at venue j. In a symmetric m-venue model this simplifies to

$$\Delta\Pi^{(m)}_{\text{gross}} = N \kappa^{(m)} \bar{\Delta}^{(m)}.$$

As with the two-venue case, the central question is whether the product of event rate, correctness gain, and penalty exceeds annualized deployment cost.

16 Factor-Model Calibration

16.1 Why Factor Calibration Matters

A company will ask how the latent event E_t or Θ_t should be estimated. The answer should not be "3by intuition."4 It should be linked to existing research infrastructure. Most quant firms already maintain factor models for market states, volatility, sector stress, ETF flows, or cross-asset pressure. The coordination strategy should therefore be calibrated against an event model that sits naturally within that framework.

16.2 Latent Factor Specification

Let

$$F_t = \begin{bmatrix} F_t^{\text{mkt}} \\ F_t^{\text{vol}} \\ F_t^{\text{flow}} \\ F_t^{\text{sector}} \end{bmatrix} \in \mathbb{R}^4,$$

with factors representing broad-market returns, realized volatility pressure, ETF or basket flow intensity, and sector-specific stress. Let the latent event score be

$$\Theta_t = \beta_0 + \beta^{\mathsf{T}} F_t + \eta_t.$$

Venue-specific signal scores are then

$$S_{j,t} = \alpha_j + \gamma_j \Theta_t + \delta_j^{\mathsf{T}} Z_{j,t} + \varepsilon_{j,t},$$

where $Z_{j,t}$ are local microstructure features such as queue imbalance, depletion rates, recent trade signs, hidden-liquidity proxies, and spread changes.

16.3 Binary Reduction

The binary signal is

$$I_{j,t} = \mathbf{1}\{S_{j,t} > \tau_j\}.$$

The posterior event probability can be estimated by logistic regression, probit regression, or a nonparametric calibration map:

$$\hat{q}(i_1, i_2) = \widehat{\mathbb{P}}(E_t = 1 \mid I_{1,t} = i_1, I_{2,t} = i_2).$$

This estimate feeds directly into the posterior-threshold rule from Section 4.

16.4 Calibrating $\bar{\Delta}$

The average coordination-error cost can be estimated by replaying historical decision epochs and comparing realized or model-implied outcomes under correct and incorrect posture labels. A practical estimate is

$$\hat{\bar{\Delta}} = \frac{1}{M} \sum_{t \in \mathcal{T}_{\text{relevant}}} \Big(\widehat{\Pi}_t(\text{correct posture}) - \widehat{\Pi}_t(\text{wrong posture}) \Big),$$

where $\mathcal{T}_{\text{relevant}}$ is the set of decision epochs that meet the signal-screening criterion and $M = |\mathcal{T}_{\text{relevant}}|$.

17 Simulation Framework

A persuasive research report should not stop at closed-form formulas. It should also show how the strategy would be stress-tested numerically.

17.1 Event-Driven Monte Carlo

The simulation loop is:

1. Simulate common factors F_t and latent event score Θ_t.
2. Simulate venue-specific microstructure scores $S_{j,t}$ and binary signals $I_{j,t}$.
3. Compute the globally optimal posture under full-information hindsight or under a calibrated posterior threshold.
4. Apply the best classical local policy and the entanglement-assisted policy.
5. Compute one-shot P&L under both policies using calibrated markout, spread, and inventory terms.
6. Aggregate over epochs to estimate expected gain, volatility of gain, and downside scenarios.

17.2 Suggested Parametric Baseline

A useful starter model is:

$$
\begin{aligned}
F_t &\sim \mathcal{N}(0, \Sigma_F), \\
\Theta_t &= \beta^\top F_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \\
S_{j,t} &= \alpha_j + \gamma_j \Theta_t + \varepsilon_{j,t}, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon), \\
I_{j,t} &= \mathbf{1}\{S_{j,t} > \tau_j\}.
\end{aligned}
$$

One can then define event-conditioned markouts and inventory costs by simple linear or nonlinear functions of Θ_t.

17.3 Performance Metrics

The company is unlikely to be persuaded by mean profit alone. The simulation should report at least:

- annual incremental mean P&L,
- standard deviation of incremental P&L,
- incremental Sharpe ratio,
- downside tail quantiles,
- sensitivity to lower entanglement availability p_Q,
- sensitivity to higher false-positive signal rates,
- break-even implementation cost.

18 Risk-Adjusted Value and Governance

18.1 Risk-Adjusted Objective

Let annual incremental profit be Y. Define a mean-variance style objective

$$J = \mathbb{E}[Y] - \lambda \mathrm{Var}(Y),$$

with risk-aversion parameter $\lambda \geq 0$. The strategy is preferable whenever $J_Q > J_C$ or, in incremental terms, whenever

$$\mathbb{E}[Y_Q - Y_C] > \lambda\big(\mathrm{Var}(Y_Q) - \mathrm{Var}(Y_C)\big).$$

A favorable feature of the posture-layer framing is that the coordination improvement can reduce loss volatility as well as mean losses, particularly when it reduces the left tail of adverse-selection events.

18.2 Operational Controls

To make the proposal governance-ready, the report should specify that the coordination layer is subject to the same controls as any other alpha or risk module:

- hard inventory limits,
- bounded order-size caps,
- per-venue kill switches,
- global kill switch,
- fallback to classical logic on telemetry failure or entanglement unavailability,
- audit logging of posture decisions.

These controls do not weaken the case; they strengthen it by showing the strategy is deployable in a production environment.

19 Why the Company Could Rationally Care

The strongest persuasive narrative is not "3quantum is powerful,"4 but rather the following.

> The firm already loses money when its distributed engines face a no-communication coordination problem. The proposed strategy targets that exact loss channel. The quantum layer is narrow, measurable, and inserted only where the current system is provably constrained by information structure rather than by poor modeling. The adoption criterion is a threshold inequality involving quantities the firm can estimate directly.

This narrative is much more likely to survive internal scrutiny from traders, risk managers, and technology leads.

20 Numerical Illustration

It is helpful to include one calibrated-style example, not as a promise but as an illustration. Suppose:

$$\begin{aligned} N &= 2 \times 10^8 \quad \text{relevant epochs/year}, \\ \omega_Q - \omega_C &= 0.1036, \\ \bar{\Delta} &= \$0.03 \quad \text{average cost of miscoordination}, \\ K_{\text{fixed}} &= \$1{,}500{,}000, \\ k_{\text{var}} &= \$0.0005. \end{aligned}$$

Then annual gross gain is

$$\Delta\Pi_{\text{gross}} = 2 \times 10^8 \times 0.1036 \times 0.03 = \$621{,}600.$$

Annual variable cost is

$$Nk_{\text{var}} = 2 \times 10^8 \times 0.0005 = \$100{,}000,$$

so annual net value is

$$\Delta\Pi_{\text{net}} = 621{,}600 - 1{,}500{,}000 - 100{,}000 = -\$978{,}400.$$

Under these assumptions the project does *not* pay. That is not a weakness of the paper; it is a strength of the framework that it reveals when the economics are not favorable.

Now suppose instead that the relevant epochs are rarer but much more costly:

$$\begin{aligned} N &= 3 \times 10^7, \\ \bar{\Delta} &= \$0.80, \\ K_{\text{fixed}} &= \$1{,}200{,}000, \\ k_{\text{var}} &= \$0.001. \end{aligned}$$

Then annual gross gain is

$$3 \times 10^7 \times 0.1036 \times 0.80 = \$2{,}486{,}400,$$

annual variable cost is \$30,000, and annual net value is

$$2{,}486{,}400 - 1{,}200{,}000 - 30{,}000 = \$1{,}256{,}400.$$

This is precisely the sort of setting where the strategy becomes attractive: fewer but much more costly no-communication decisions.

21 Empirical Definition of Coordination Events

In order to move from a theoretical tacit coordination game to a strategy that can be implemented within a trading firm, it is necessary to formally define what constitutes a *coordination event* within the firm's execution environment.

21.1 Local Signal Structure

Let trading venue $i \in \{1, 2\}$ produce a local signal

$$s_i(t) \in \{-1, +1\}$$

where

- $s_i(t) = +1$ indicates upward price pressure
- $s_i(t) = -1$ indicates downward price pressure

Signals are generated from observable microstructure features:

$$s_i(t) = \text{sign}\left(\alpha_1 \Delta\text{LOB}_i(t) + \alpha_2 \text{OFI}_i(t) + \alpha_3 \text{TradeImbalance}_i(t) + \alpha_4 \text{ShortTermMomentum}_i(t)\right)$$

where

- ΔLOB = order book imbalance
- OFI = order flow imbalance
- trade imbalance = buyer/seller initiated trades

21.2 Coordination Requirement

A coordination event occurs when two venue engines must take matching directional actions:

$$a_1(t) = a_2(t)$$

Failure to coordinate generates execution losses due to

- crossed inventory exposure
- adverse selection
- missed cross-venue arbitrage

21.3 Formal Event Definition

Define the event indicator

$$E_t = \begin{cases} 1 & \text{if } |s_1(t) - s_2(t)| \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where θ represents the signal disagreement tolerance.

The total number of coordination events over a trading horizon T is

$$N_E = \sum_{t=1}^{T} E_t$$

which can be empirically measured from historical market data.

22 Economic Cost of Miscoordination

The economic impact of failed coordination is measurable using historical data.

22.1 Execution Loss Model

Let

$$a_i(t) \in \{-1, +1\}$$

be the action taken by venue engine i.

Miscoordination occurs when

$$a_1(t) \neq a_2(t)$$

The associated loss is

$$L_t = Q_t \cdot |P_2(t + \Delta) - P_1(t + \Delta)|$$

where

- Q_t = traded size
- Δ = settlement horizon

22.2 Expected Loss

The expected loss per event is

$$\mathbb{E}[L] = \mathbb{E}[Q] \cdot \mathbb{E}[|P_2 - P_1|]$$

Total expected loss:

$$L_{\text{total}} = N_E \cdot \mathbb{E}[L]$$

This quantity represents the upper bound on the value of improved coordination.

23 Classical Baseline Coordination Performance

Before evaluating any quantum-assisted coordination protocol, the best classical performance must be measured.

23.1 Shared Randomness Strategy

The optimal classical strategy under no communication is

$$a_i = \text{sign}(s_i \oplus r)$$

where r is shared randomness.

The maximal classical success probability for XOR-type coordination games is

$$p_C = 0.75$$

23.2 Observed Desk Performance

Let

$$p_{\text{desk}}$$

represent the empirical coordination rate of the firm's current infrastructure.

The effective improvement potential is therefore

$$\Delta p = p_Q - p_{\text{desk}}$$

where

$$p_Q = \cos^2(\pi/8) \approx 0.8536$$

is the theoretical quantum success rate.

24 Profitability Model

The annual economic gain from improved coordination is

$$G = N_E \cdot \Delta p \cdot \mathbb{E}[L]$$

where

- N_E = coordination events per year
- Δp = improvement in coordination probability
- $\mathbb{E}[L]$ = expected loss per miscoordination

24.1 Break-Even Condition

Let C_{deploy} denote the annualized system cost.

Deployment is profitable when

$$G > C_{\text{deploy}}$$

or equivalently

$$N_E \cdot \Delta p \cdot \mathbb{E}[L] > C_{\text{deploy}}$$

25 Operational Feasibility at Production Latency

A common objection to quantum-assisted coordination is that quantum systems cannot operate at high-frequency trading latency.

25.1 Latency Decomposition

Total system latency:

$$T_{\text{total}} = T_{\text{signal}} + T_{\text{measurement}} + T_{\text{decision}}$$

Typical values:

- signal extraction: 100–300 ns
- quantum measurement: ~ 10 ns
- decision logic: 50–100 ns

Thus

$$T_{\text{total}} \approx 200 - 400 \text{ ns}$$

which is comparable to modern FPGA-based trading pipelines.

26 Mapping XOR Coordination Games to Trading Decisions

The CHSH/XOR game abstraction corresponds directly to binary trading decisions.

26.1 Game Variables

$$x_i \in \{0, 1\}$$

represents the local market state.

$$a_i \in \{0, 1\}$$

represents the trading decision.

The coordination rule is

$$a_1 \oplus a_2 = x_1 \wedge x_2$$

26.2 Trading Interpretation

- $x_i = 1$: toxic flow detected
- $x_i = 0$: normal flow

The optimal joint policy is therefore

- hedge jointly when both venues detect toxicity
- continue liquidity provision otherwise

This binary structure matches real desk decisions such as

- widen spread
- pull liquidity
- hedge inventory

27 Empirical Backtesting Framework

The strategy can be validated using historical market data.

27.1 Backtest Steps

1. Reconstruct venue-level signals $s_1(t), s_2(t)$
2. Identify coordination events E_t
3. Simulate classical coordination policy
4. Simulate quantum-improved coordination probability
5. Measure resulting PnL difference

The resulting profit estimate provides a firm-specific evaluation of the strategy.

28 Empirical Calibration Using Historical Market Data

To evaluate firm-specific profitability, the model parameters must be calibrated using historical market data.

28.1 Data Sources

Calibration requires the following datasets:

- Full depth limit order book data
- Trade prints with aggressor flags
- Venue-specific timestamps
- Quote update messages

These datasets allow reconstruction of high-frequency market states across multiple venues.

28.2 Signal Estimation

Local venue signals are computed as

$$s_i(t) = \text{sign}\left(\beta_1 \cdot \text{OFI}_i(t) + \beta_2 \cdot \text{LOBImbalance}_i(t) + \beta_3 \cdot \text{ShortTermReturn}_i(t)\right)$$

where

- OFI = order flow imbalance
- LOBImbalance = depth imbalance
- ShortTermReturn = short-horizon price movement

Parameters β are estimated using logistic regression or regularized classification models.

28.3 Event Frequency Estimation

Coordination events are identified by

$$E_t = \mathbf{1}\left(|s_1(t) - s_2(t)| \leq \theta\right)$$

The annual event frequency is then

$$N_E = \sum_{t=1}^{T} E_t$$

where T represents the total number of trading intervals.

This produces a firm-specific estimate of coordination opportunities.

29 Monte Carlo Microstructure Simulation

To evaluate the economic impact of improved coordination, we simulate cross-venue market dynamics using a stochastic microstructure model.

29.1 Price Dynamics

Prices follow a discrete-time diffusion:

$$P_{t+1} = P_t + \sigma \epsilon_t$$

where

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

and σ represents short-horizon volatility.

29.2 Order Flow Model

Trade arrivals follow a Poisson process:

$$N(t) \sim \text{Poisson}(\lambda t)$$

with aggressor direction determined by a Bernoulli process.

29.3 Coordination Policies

We compare two policies:

1. Classical coordination strategy

$$p_C = 0.75$$

2. Quantum-assisted coordination strategy

$$p_Q = \cos^2(\pi/8)$$

29.4 Simulation Output

For each simulation run we compute:

- total PnL
- coordination success rate

- adverse selection losses

The difference in expected PnL between policies provides an estimate of the economic value of improved coordination.

30 Multi-Venue Coordination Extension

Real trading environments involve more than two venues.

Let

$$i \in \{1, 2, \ldots, V\}$$

denote trading venues.

Each venue produces a local signal

$$s_i \in \{-1, 1\}$$

and an action

$$a_i \in \{-1, 1\}$$

The coordination objective becomes

$$a_1 = a_2 = \cdots = a_V$$

subject to no communication between engines.

30.1 Generalized Coordination Probability

If pairwise coordination success probability is p, then the probability of full coordination across V venues is

$$P_{coord} = p^{V-1}$$

Quantum-assisted correlations therefore become increasingly valuable as the number of venues increases.

30.2 Economic Scaling

The expected economic gain scales as

$$G_V = N_E^{(V)} \cdot \Delta p^{(V)} \cdot \mathbb{E}[L]$$

where

- $N_E^{(V)}$ = coordination events across V venues
- $\Delta p^{(V)}$ = improvement in coordination probability

This scaling property suggests that multi-venue market making is particularly well suited for coordination-enhancing strategies.

31 What a Bulletproof Version Can and Cannot Mean

No honest technical paper should promise that a strategy is perfect, flawless, or guaranteed to make money in all environments. What a rigorous paper *can* do is derive a set of sufficient conditions under which the strategy is profitable and beneficial. That is the mathematically defensible meaning of "3bulletproof"4 in a serious research setting.

Accordingly, this document should be read as proving the following conditional statement:

> If the firm's latency-constrained decision problem is well approximated by a tacit coordination game with a positive quantum-classical value gap, and if the average coordination-error loss exceeds the break-even threshold implied by implementation cost, then the proposed coordination layer strictly improves expected annual profit.

That statement is strong, actionable, and testable.

32 Expanded Theoretical Results

32.1 Nonuniform Input Distribution

The uniform CHSH benchmark is the cleanest analytical starting point, but the firm's real signal distribution is generally nonuniform. Let

$$\pi_{ab} = \mathbb{P}(I_1 = a, I_2 = b), \qquad a, b \in \{0, 1\}.$$

Suppose the payoff is one when the parity rule is satisfied and zero otherwise. Then the classical value solves a weighted combinatorial optimization over deterministic local rules. Because there are only sixteen deterministic pairs (f_1, f_2) in the binary-binary case, the classical value can be computed exactly by enumeration:

$$\omega_C(\pi) = \max_{f_1, f_2} \sum_{a,b \in \{0,1\}} \pi_{ab} \mathbf{1}\{f_1(a) \oplus f_2(b) = 1 - ab\}.$$

The quantum value can be characterized as the optimum over observables with outcomes ± 1 acting on a shared state. In practical work, one can estimate both values numerically after calibrating π.

32.2 State-Weighted Optimization

If the dollar gain from solving the double-confirmation state is the dominant economic term, a more relevant objective is the state-weighted value

$$V(\sigma) = \sum_{a,b} \pi_{ab} \Delta_{ab} s_\sigma(a,b),$$

where $s_\sigma(a,b)$ is the correctness probability under strategy σ. This formulation makes clear that the firm does not need a generic quantum advantage on every state; it needs improved coordination where the dollars are.

32.3 Approximate Reduction Error

In realistic settings the globally optimal posture rule need not be exactly Boolean. Let $\widetilde{P}^*$ be the true optimal posture map and P^* its reduced Boolean approximation. Define the reduction error

$$\varepsilon_{\text{red}} = \mathbb{E}\big[\Pi(\widetilde{P}^*, E) - \Pi(P^*, E)\big].$$

Then the net benefit of using the quantum-assisted reduced game instead of the best classical reduced policy is bounded below by

$$\Delta\Pi_{\text{gross}} - \varepsilon_{\text{red}}.$$

Thus the reduction remains justified whenever the coordination gain exceeds the modeling loss from Boolean approximation.

33 A Dynamic Inventory-Control Extension

Inventory enters naturally because posture choice affects fill rates and net position. Let $X_t \in \mathbb{R}$ denote net inventory and suppose

$$X_{t+1} = X_t + U_t + \xi_t,$$

where U_t is a deterministic function of the chosen posture and realized fills, and ξ_t is exogenous noise. Let the stage reward be

$$r_t = \Pi_t - \lambda_X X_t^2.$$

The firm wants to maximize

$$V_0 = \mathbb{E}\left[\sum_{t=0}^{T-1} \delta^t r_t\right].$$

If the entanglement-assisted layer reduces the probability of entering the wrong posture on high-risk epochs, then it decreases both immediate adverse-selection losses and expected future inventory penalties. In other words, the one-shot coordination edge has a dynamic amplification channel through state evolution.

Proposition 33.1 (Dynamic amplification via inventory)**.** *Suppose the wrong posture increases the conditional mean-square next-period inventory by at least $\kappa > 0$ on a set of relevant epochs*

of probability mass p. Then the dynamic value of improving correctness by Δs per relevant epoch exceeds the static one-shot gain by at least

$$\delta \lambda_X p \kappa \Delta s.$$

Proof. Compare the Bellman recursion under the two policies and use the lower bound on the incremental expected quadratic inventory term. □

34 A Three-Posture Generalization

Binary posture is analytically elegant, but traders often think in three postures:

1. **Passive/normal,**
2. **Protective/widen-and-cancel,**
3. **Directional/aggressive.**

A useful extension is to let the local actions remain binary while the joint action map generates a ternary posture. For example,

$$(A_1, A_2) \mapsto \begin{cases} \text{Passive}, & (0,1), \\ \text{Protective}, & (1,0), \\ \text{Directional}, & (0,0) \text{ or } (1,1), \end{cases}$$

after appropriate relabeling. The reduction to a nonlocal game becomes more nuanced, but the same principle holds: the company only needs the local actions to encode a small posture variable, not a full trade schedule.

35 Sensitivity Tables to Include in a Final Memo

A production-quality version of this paper should include tables such as:

- net annual value as a function of N and $\bar{\Delta}$,
- break-even $\bar{\Delta}$ as a function of fixed cost and availability p_Q,
- sensitivity to false-positive signal rates,
- sensitivity to event rarity,
- sensitivity to dynamic inventory penalty λ_X.

These can be produced directly from the formulas above and from the simulation framework.

36 Model-Risk Appendix Summary

The company will appreciate an explicit acknowledgment of model risk together with a mitigation plan.

- **Signal-model risk:** mitigate via recalibration and regime segmentation.
- **Cost-model risk:** mitigate via conservative markout and inventory estimates.
- **Availability risk:** mitigate via classical fallback.
- **Operational risk:** mitigate via bounded posture authority and kill switches.

The existence of model risk does not weaken the argument; the lack of a mitigation plan would.

37 Conclusion

The main accomplishment of this document is not that it waves at Bell inequalities in a financial context. It is that it builds a concrete, trader-relevant, mathematically defensible case for a specific use of entanglement-assisted coordination inside a latency-constrained trading system.

The strategy becomes compelling when three conditions hold:

1. the firm's decision problem is genuinely decentralized and time-constrained,
2. the average cost of miscoordination is large enough,
3. the quantum-assisted coordination layer can be deployed with bounded annual cost and robust fallback controls.

When those conditions hold, the strategy is not a science experiment. It is a capital-allocation decision governed by a clean threshold inequality.

38 Semidefinite Optimization View of the Quantum Problem

A practical advantage of XOR-type coordination games is that the quantum value admits efficient numerical approximation through semidefinite programming. In the two-party binary-output setting one may write the correlation matrix of observables and optimize over positive semidefinite moment matrices subject to normalization constraints. This matters for company persuasion because it provides a reproducible computational pipeline rather than a purely symbolic argument.

Let X_i and Y_j denote centered observables associated with local measurement choices. The expected game score can be expressed as a linear form in pairwise correlations $\mathbb{E}[X_i Y_j]$. The admissible quantum set is larger than the classical correlation polytope but is still tractable through semidefinite relaxations. Therefore, once the firm specifies a weighted payoff matrix, the corresponding quantum benchmark can be computed and stress-tested numerically.

38.1 Weighted XOR Objective

Suppose the reduced payoff uses state weights c_{ab} and sign matrix $M_{ab} \in \{-1, +1\}$. Then the score can be written as

$$S = \sum_{a,b} c_{ab} M_{ab} \mathbb{E}[X_a Y_b].$$

The classical benchmark is the maximum over $x_a, y_b \in \{-1, +1\}$ of the same linear form, while the quantum benchmark is the maximum over unit vectors u_a, v_b of

$$S_Q = \sum_{a,b} c_{ab} M_{ab} \langle u_a, v_b \rangle.$$

By Tsirelson's theorem this vector formulation is exact for the binary-output XOR setting. In calibration work, this permits direct evaluation of the incremental value under nonuniform state weights.

38.2 Why This Helps Internally

Inside a quant firm, the semidefinite-programming formulation serves two purposes. First, it lets research verify that the reduced game actually has a nontrivial quantum-classical gap after inserting empirically estimated state weights. Second, it avoids over-reliance on the symmetric CHSH benchmark. The company can say: "we did not just inherit a textbook gap; we recomputed the gap for our own weighted decision problem."

39 Signal Engineering for a Trading Desk

A practical paper should move beyond an abstract binary signal and say what the desk can realistically compute at sub-millisecond latency. A useful local signal library contains the following classes.

39.1 Queue-Pressure Signals

For venue j let $B_{j,t}$ and $A_{j,t}$ denote best-bid and best-ask queue sizes, and let $\Delta B_{j,t}$ and $\Delta A_{j,t}$ denote short-window changes. Examples of local features include:

$$\begin{aligned}
\mathrm{Imb}_{j,t} &= \frac{B_{j,t} - A_{j,t}}{B_{j,t} + A_{j,t}}, \\
\mathrm{Dep}_{j,t} &= -\min\{\Delta B_{j,t}, \Delta A_{j,t}\}, \\
\mathrm{Thin}_{j,t} &= \mathbf{1}\{B_{j,t} + A_{j,t} < \vartheta_j\}.
\end{aligned}$$

A high toxicity score may be triggered when imbalance, depletion, and thin-book indicators jointly exceed a calibrated threshold.

39.2 Trade-Flow Signals

Let $N^+_{j,t}(h)$ and $N^-_{j,t}(h)$ denote counts of buyer- and seller-initiated trades on a short horizon h. One may define signed trade-flow intensity

$$\mathrm{Flow}_{j,t}(h) = \frac{N^+_{j,t}(h) - N^-_{j,t}(h)}{N^+_{j,t}(h) + N^-_{j,t}(h) + 1}.$$

Aggressive bursts, especially when paired with queue depletion, are natural ingredients in a toxicity warning.

39.3 Crossed-Book and Fragmentation Signals

In fragmented markets the same symbol may have partial pressure distributed across venues. Let m_t denote a national reference mid, and define venue-specific dislocation

$$\mathrm{Disloc}_{j,t} = \frac{m_{j,t} - m_t}{m_t}.$$

Persistent dislocations, especially when aligned with local aggressive flow, may help distinguish harmless local noise from a true common latent event.

39.4 Event Compression into a Binary Signal

For each venue use a local score

$$S_{j,t} = w_j^{\mathsf{T}} Z_{j,t},$$

where $Z_{j,t}$ contains standardized features and w_j are trained coefficients. The binary signal is then

$$I_{j,t} = \mathbf{1}\{S_{j,t} > \tau_j\}.$$

This keeps the coordination layer simple while permitting a rich underlying feature set.

40 Market-Impact and Execution-Cost Extension

The earlier sections model posture errors through markouts, inventory, and missed alpha. A more complete version should also address market impact.

40.1 Temporary Impact

Suppose that choosing the defensive or directional posture generates an expected temporary impact cost

$$C_{\mathrm{tmp}}(q) = \phi|q| + \psi q^2,$$

for response size q. The expected incremental benefit of switching posture under event probability $q(i_1, i_2)$ then becomes

$$B(i_1, i_2) = q(i_1, i_2)\pi_D - (1 - q(i_1, i_2))\ell_D - C_{\text{tmp}}(q^*).$$

This simply alters the posterior threshold and therefore fits seamlessly into the same mathematical structure.

40.2 Permanent Impact and Strategic Selectivity

If the firm's response trades carry informational footprint or expected permanent impact, then the coordination layer should be even more selective. This strengthens the value of solving the decentralized confirmation problem accurately, because the cost of unnecessary switching rises.

41 Portfolio-Level Extension

Firms rarely trade a single symbol in isolation. A stronger version of the proposal allows the latent event to affect a portfolio or basket.

41.1 Vector Inventory State

Let $X_t \in \mathbb{R}^d$ be inventory across d related instruments. Let the wrong posture create an expected inventory shock $G(I_t, A_t)$, so that

$$X_{t+1} = X_t + G(I_t, A_t) + \xi_t.$$

The quadratic risk penalty is

$$X_t^\top \Lambda X_t,$$

with factor-structured matrix

$$\Lambda = B\Gamma B^\top + D,$$

where B contains factor loadings, Γ is a factor-risk matrix, and D is diagonal idiosyncratic cost. This provides a clean bridge from the coordination problem into the firm's existing portfolio-risk infrastructure.

41.2 Basket Event Example

Consider a semiconductor basket where local venue warnings on one or two constituents may reflect basket-level pressure induced by ETF or options hedging flows. In such settings the latent event is better understood at the portfolio level than at the single-symbol level. The coordination layer can still be binary, but the cost of getting the joint posture wrong may be much larger because the inventory dislocation propagates across names.

42 Backtest Design for a Production Research Pipeline

The desk will care a great deal about how backtesting is done. An imprecise backtest undermines an otherwise elegant idea.

42.1 Event Definition

A relevant decision epoch should satisfy all of the following:

1. at least one venue emits a local warning,
2. the symbol or basket is tradable under the firm's normal operational filters,
3. the decision horizon is shorter than or comparable to the inter-venue communication delay that motivates tacit coordination,
4. sufficient downstream market data exist to estimate markout and inventory effects.

Restricting to these epochs avoids inflating N with irrelevant observations.

42.2 Counterfactual Labeling

For each epoch, build three labels:

- **Correct posture label**: the ex post or model-implied best posture,
- **Best classical local label**: posture induced by the best admissible classical policy,
- **Quantum-assisted label**: posture induced by the chosen quantum strategy or its simulated analogue.

The objective is not to compare against an unrealistic strawman; it is to compare against the best local-information benchmark that respects the no-communication constraint.

42.3 Avoiding Look-Ahead Bias

Any estimation of markouts, event probabilities, or factor states must be done with information available at the decision time. This sounds obvious, but in fragmented market data it is easy to leak future information inadvertently through synchronization choices. A robust internal report should explicitly state how timestamps are normalized and how late packets are handled.

42.4 Cross-Validation by Regime

A serious paper should not rely on one pooled sample. It should report performance across market regimes such as:

- high-volatility days,
- low-volatility days,

- macro-event windows,
- ETF rebalance periods,
- earnings or news clusters,
- open, midday, and close subsamples.

The strategy is most likely to shine exactly when coordination-error costs are largest, which may coincide with turbulent regimes.

43 Scenario Analysis and Stress Tests

A company will ask what happens if assumptions deteriorate. The paper should therefore state the key stress dimensions.

43.1 Lower Entanglement Availability

If the hybrid availability p_Q is small, value declines linearly:

$$\Delta\Pi_{\text{hyb,gross}} = N p_Q(\omega_Q - \omega_C)\bar{\Delta}.$$

This is favorable from a risk perspective because the downside of lower availability is graceful degradation rather than catastrophic failure.

43.2 Signal Deterioration

If local features lose predictive power, the posterior threshold may be crossed less accurately, reducing $\bar{\Delta}$ as measured on the screened event set. This can be monitored through routine feature-drift and markout-drift dashboards.

43.3 Higher Cost of Switching

If defensive or directional posture changes become more expensive, for example because of exchange fee changes or worsening impact, the threshold τ rises. This does not break the framework; it simply means the coordination layer should be invoked on a narrower subset of more severe events.

43.4 Latency Compression by Classical Infrastructure

If classical communication improves dramatically, the universe of truly no-communication decisions shrinks. Again, this does not invalidate the mathematics; it changes N. The strategy remains attractive only if enough costly decentralized events survive.

44 A Research Roadmap for the Company

One reason proposals fail internally is that they jump directly from theory to production. The company is more likely to listen if the deployment path is staged.

44.1 Stage 1: Offline Quant Research

This stage estimates the event distribution, the wrong-action penalty, and the weighted game values using historical data. No production changes are required. The outcome is a calibrated estimate of break-even economics.

44.2 Stage 2: Historical Replay with Simulated Coordination Oracle

Before any real quantum hardware is considered, the desk can replay the event stream using a simulated coordination oracle that enforces the theoretically optimal correlation pattern. This isolates the business value of improved coordination from hardware questions.

44.3 Stage 3: Shadow Mode

If Stage 2 looks promising, the desk can run the posture layer in shadow mode next to the production stack. This allows validation of operational telemetry, event rates, and risk-control interactions without affecting live trading.

44.4 Stage 4: Bounded-Risk Pilot

A narrowly bounded-risk strategy may then use the coordination layer only on a small subset of symbols or under a very tight response-size cap. This ensures that empirical evaluation is feasible without exposing the firm to uncontrolled implementation risk.

45 Communication Strategy for Internal Stakeholders

Different stakeholders will evaluate the proposal differently.

45.1 For Traders

Emphasize that the layer aims to reduce costly posture mistakes under sweep risk and fragmentation, not to replace trader intuition. The language should focus on markouts, fill quality, and missed opportunities.

45.2 For Quant Researchers

Emphasize the Bayesian team-game formulation, exact break-even inequality, and reproducible optimization of weighted quantum-classical values. Quant researchers will care about whether the gain survives realistic state weights and nonuniform signal distributions.

45.3 For Technology Leads

Emphasize the hybrid architecture, bounded scope, fallback logic, and telemetry requirements. The technology team will care less about Bell inequalities than about interfaces, latency budgets, and failure handling.

45.4 For Risk and Compliance

Emphasize that the coordination layer only sets a compact posture variable and is embedded inside the existing risk envelope. Compliance does not need to accept mysterious autonomous behavior; it only needs to approve a new signal-to-posture map with full logging and kill switches.

46 Further Mathematical Appendices

46.1 A Concentration Bound for Estimated Value

Suppose the backtest yields i.i.d. or weakly dependent epoch-level incremental gains $Y_1, \ldots, Y_M$ bounded in $[a, b]$. Then Hoeffding-style concentration gives

$$\mathbb{P}\left(\left| \frac{1}{M} \sum_{t=1}^{M} Y_t - \mathbb{E}[Y] \right| > \epsilon \right) \leq 2 \exp\left(-\frac{2M\epsilon^2}{(b-a)^2} \right).$$

This can be used to size the sample required for a statistically convincing offline study.

46.2 A Conservative Lower Bound on Net Value

If $\underline{p}_Q$ is a lower confidence bound for availability and $\underline{\Delta}$ a lower confidence bound for average miscoordination loss, then a conservative lower bound on annual net value is

$$\underline{\Delta\Pi}_{\text{net}} = N\underline{p}_Q(\omega_Q - \omega_C)\underline{\Delta} - K_{\text{fixed}} - Nk_{\text{var}}.$$

This is the quantity a cautious committee should compare with zero.

46.3 Model Averaging Across Regimes

Suppose the market alternates among regimes $r \in \{1, \ldots, R\}$ with probabilities λ_r. Then total annual net value is

$$\Delta\Pi_{\text{net}} = \sum_{r=1}^{R} \lambda_r N_r p_{Q,r}(\omega_{Q,r} - \omega_{C,r})\bar{\Delta}_r - K_{\text{fixed}} - \sum_{r=1}^{R} \lambda_r N_r k_{\text{var},r}.$$

This makes explicit that the project can be attractive even if most of the value comes from a small number of high-stress regimes.

47 Extended Conclusion

A company does not need a proof that quantum coordination is universally superior in all possible trading problems. It needs a credible proof that, for a specific and costly class of decentralized latency-bound decisions, improved coordination can be monetized. This paper provides that proof structure.

The central chain is now complete:

1. start from a trader-relevant event class,
2. derive a Bayesian posterior-threshold rule,
3. reduce the joint-decision bottleneck to a weighted coordination game,
4. compute the classical and quantum values,
5. map the value gap into dollars through calibrated wrong-action penalties,
6. compare that gain with bounded implementation cost,
7. validate the economics through backtest and simulation,
8. and deploy, if at all, only as a narrow hybrid coordination layer with classical fallback.

That is the strongest mathematically serious, economically grounded, and operationally defensible form of the proposal.

48 Additional Calibration and Validation Considerations

48.1 Estimating the Relevant Event Rate

The annual event count N should not be taken as the number of all market-data updates or all order decisions. It should be the number of *economically relevant decentralized epochs* in which the wrong joint posture has a measurable cost and communication is not fast enough to resolve the decision. A conservative estimator is

$$\hat{N} = \sum_{t=1}^{T} \mathbf{1}\{\max_j I_{j,t} = 1\}\mathbf{1}\{\text{communication-infeasible at } t\}\mathbf{1}\{\text{symbol passes operational filters}\}.$$

This estimator automatically excludes benign periods and prevents overstatement of business value.

48.2 Out-of-Sample Penalty Stability

One of the most important internal validation exercises is to check whether $\hat{\bar{\Delta}}$ is stable out of sample. Let the sample be partitioned into folds $\mathcal{T}_1, \ldots, \mathcal{T}_K$. Define

$$\hat{\bar{\Delta}}_k = \frac{1}{|\mathcal{T}_k|} \sum_{t \in \mathcal{T}_k} \Big(\widehat{\Pi}_t(\text{correct}) - \widehat{\Pi}_t(\text{wrong}) \Big).$$

The report should include the dispersion of $\hat{\bar{\Delta}}_k$ across folds. If the mean is positive but the dispersion is very large, the proposal should be presented as regime-dependent rather than universally attractive.

48.3 Conservative Event Filtering

The company is more likely to approve a pilot if the initial research filters aggressively toward high-cost events. Let $\mathcal{E}_\kappa$ be the set of epochs with estimated wrong-action penalty above the κth quantile. Then one can focus on

$$\bar{\Delta}^{(\kappa)} = \mathbb{E}[\Delta_t \mid t \in \mathcal{E}_\kappa].$$

As κ increases, N falls but $\bar{\Delta}^{(\kappa)}$ rises. This creates a research frontier from which the company can choose a conservative deployment region.

49 Detailed Dynamic-Programming Interpretation

The decentralized posture layer can be embedded inside a standard dynamic-programming architecture. Let the complete state be

$$\mathcal{S}_t = (X_t, F_t, \mathcal{M}_t),$$

where X_t is inventory, F_t is a factor state, and $\mathcal{M}_t$ collects local order-book summaries. The centralized Bellman equation under full communication is

$$V_t(\mathcal{S}_t) = \max_{P_t \in \{H,D\}} \mathbb{E}\big[r_t(\mathcal{S}_t, P_t) + \delta V_{t+1}(\mathcal{S}_{t+1}) \mid \mathcal{S}_t \big].$$

Under latency constraints the actual system cannot implement that centralized policy at decision time. Instead it must approximate the Bellman-implied posture decision using local maps. The value of entanglement-assisted coordination is therefore the value of a better decentralized approximation to the centralized policy.

49.1 Approximation Gap

Let V_t^{cent} denote the centralized value, V_t^C the best classical local value, and V_t^Q the entanglement-assisted local value. Then the relevant inequalities are

$$V_t^{\text{cent}} \geq V_t^Q \geq V_t^C.$$

The first inequality reflects the fact that even quantum coordination does not create post-observation communication. The second is the nonlocal advantage. This framing is useful internally because it places the proposal inside a familiar control-theoretic hierarchy.

50 Suggested Tables for the Final Submission

The following tables should be included after calibration, even though the current document keeps them conceptual.

50.1 Table 1: Event Statistics

Columns should include symbol group, venue pair, annual event count N, mean and median estimated Δ_t, and communication-feasible versus communication-infeasible split.

50.2 Table 2: Policy Comparison

Columns should include baseline classical policy, predictive classical policy, simulated quantum-assisted policy, mean incremental P&L, standard deviation, lower-tail quantiles, and turnover or posture-switch counts.

50.3 Table 3: Break-Even Cost Frontier

Rows should vary p_Q and columns should vary conservative values of $\bar{\Delta}$, reporting the maximum fixed annual cost consistent with positive net value:

$$K_{\max} = N p_Q(\omega_Q - \omega_C)\bar{\Delta} - N k_{\text{var}}.$$

This is precisely the table senior management will understand.

51 Final Managerial Framing

The most persuasive closing line for a company is not that the strategy is magical; it is that the company already pays a decentralization tax whenever communication is too slow. This proposal is a mathematically explicit attempt to reduce that tax. If offline calibration shows the tax is small, the project should be rejected. If calibration shows the tax is large, then the break-even inequality tells the firm exactly how much implementation it can afford.

A Notation Summary

Symbol	Meaning
E_t	Latent event indicating genuine short-horizon toxicity or directional pressure
Θ_t	Continuous latent event score
$I_{j,t}$	Binary local signal observed at venue j
$A_{j,t}$	Local action bit chosen by venue j
P_t	Firm-level posture
π_D, π_H	Gains from correct directional and correct hedged posture
ℓ_D, ℓ_H	Losses from false directional and false hedged posture
$\bar{\Delta}$	Average dollar cost of a coordination error
N	Number of relevant decision epochs per year
ω_C, ω_Q	Optimal classical and quantum correctness values
K_{fixed}	Annualized fixed implementation cost
k_{var}	Variable cost per decision epoch
p_Q	Probability that quantum coordination is available
X_t	Inventory state
Λ	Inventory-cost matrix

B Exact Enumeration of Deterministic Binary Local Policies

In the binary-binary two-player setting there are four deterministic local rules for each player:

$$0, \quad 1, \quad x, \quad 1-x,$$

where $x \in \{0,1\}$ is the local signal. Hence there are sixteen pairs (f_1, f_2). For any nonuniform input distribution π, the classical value can be computed exactly by enumerating these sixteen pairs and selecting the best weighted score. This is useful both for theory and for code validation in simulation.

C Proof of the Break-Even Condition

Starting from

$$\Delta\Pi_{\text{net}} = N(\omega_Q - \omega_C)\bar{\Delta} - K_{\text{fixed}} - Nk_{\text{var}},$$

profitability requires $\Delta\Pi_{\text{net}} > 0$. Rearranging gives

$$\bar{\Delta} > \frac{K_{\text{fixed}} + Nk_{\text{var}}}{N(\omega_Q - \omega_C)} = \frac{K_{\text{fixed}}/N + k_{\text{var}}}{\omega_Q - \omega_C}.$$

No further assumptions are needed.

D Pseudo-Code for Historical Backtest Calibration

```
Input: Historical venue-level event stream, markout model, inventory cost model
Output: Estimated N, Delta_bar, and policy-comparison statistics

1. Build synchronized event epochs across venues.
2. Compute local microstructure features Z_{j,t}.
3. Convert features to local scores S_{j,t} and binary signals I_{j,t}.
4. Estimate posterior event probabilities q(i1,i2).
5. Define globally optimal posture labels using the calibrated threshold rule.
6. Compute realized or modeled P&L under:
   (a) best classical local policy,
   (b) entanglement-assisted policy,
   (c) strong predictive classical baseline.
7. Estimate Delta_bar from wrong-vs-right posture comparisons.
8. Compute annualized net value using deployment-cost assumptions.
9. Run sensitivity analysis over thresholds, costs, and event frequencies.
```

E Suggested Overleaf Compilation Notes

This document is intentionally self-contained. It does not require BibTeX or external figures to compile. If the user wants to split it into files later, a natural structure is:

- main.tex
- sections/theory.tex
- sections/economics.tex
- sections/implementation.tex
- sections/appendix.tex

For the current version, a single-file workflow is simplest and least error-prone.

References

[1] Dawei Ding and Liang Jiang, *Coordinating Decisions via Quantum Telepathy*, arXiv:2407.21723, 2024.

[2] J. S. Bell, *On the Einstein Podolsky Rosen Paradox*, Physics, 1(3):195–200, 1964.

[3] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, *Proposed Experiment to Test Local Hidden-Variable Theories*, Physical Review Letters, 23(15):880–884, 1969.

[4] B. S. Cirel'son, *Quantum Generalizations of Bell's Inequality*, Letters in Mathematical Physics, 4:93–100, 1980.

[5] Gilles Brassard, Anne Broadbent, and Alain Tapp, *Quantum Pseudo-Telepathy*, Foundations of Physics, 35:1877–1907, 2005.

[6] John C. Harsanyi, *Games with Incomplete Information Played by "Bayesian" Players, I–III*, Management Science, 1967–1968.

[7] Roy Radner, *Team Decision Problems*, Annals of Mathematical Statistics, 33(3):857–881, 1962.

[8] Jacob Marschak, *Elements for a Theory of Teams*, Management Science, 1(2):127–137, 1955.

[9] Maureen O'Hara, *Market Microstructure Theory*, Blackwell, 1995.

[10] Joel Hasbrouck, *Empirical Market Microstructure*, Oxford University Press, 2007.

[11] Álvaro Cartea, Sebastian Jaimungal, and José Penalva, *Algorithmic and High-Frequency Trading*, Cambridge University Press, 2015.

[12] Bruno Biais, Larry Glosten, and Chester Spatt, *Market Microstructure: A Survey of Microfoundations, Empirical Results, and Policy Implications*, Journal of Financial Markets, 8(2):217–264, 2005.